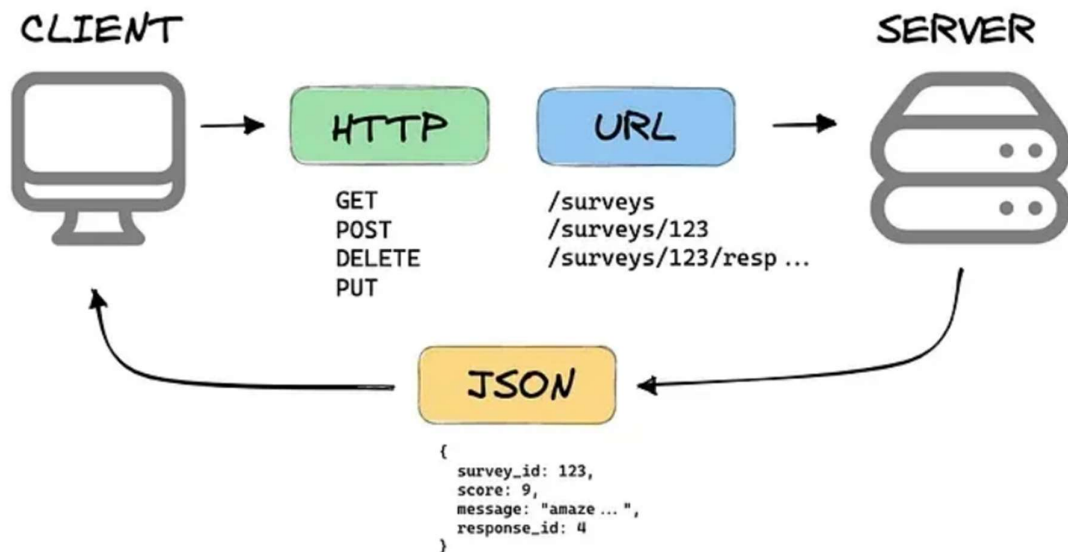


REST API

✚ What is REST API :-

- REST API (**R**epresentational **S**tate **T**ransfer **A**PI) is a way for two systems to communicate with each other over the internet using the **HTTP** protocol.

WHAT IS A REST API?



- ✚ **REST APIs allow** clients (like Postman, mobile apps, web apps) to perform actions such as:

- Saving data
- Reading data
- Updating data
- Deleting data

Client sends a request → Server sends a response in JSON format.

Key Concepts of REST API :-

1. Client–Server Architecture
2. Stateless
3. Resource-Based
4. HTTP Methods
5. JSON Format
6. Uniform Interface
7. Layered System

HTTP Methods :-

Method	Meaning	Example
GET	Read Data	GET / students
POST	Create Data	POST / students
PUT	Update full data	PUT / students / 5
PATCH	Update part of data	PATCH / students / 5
DELETE	Delete data	DELETE / students / 5

Short Summary :-

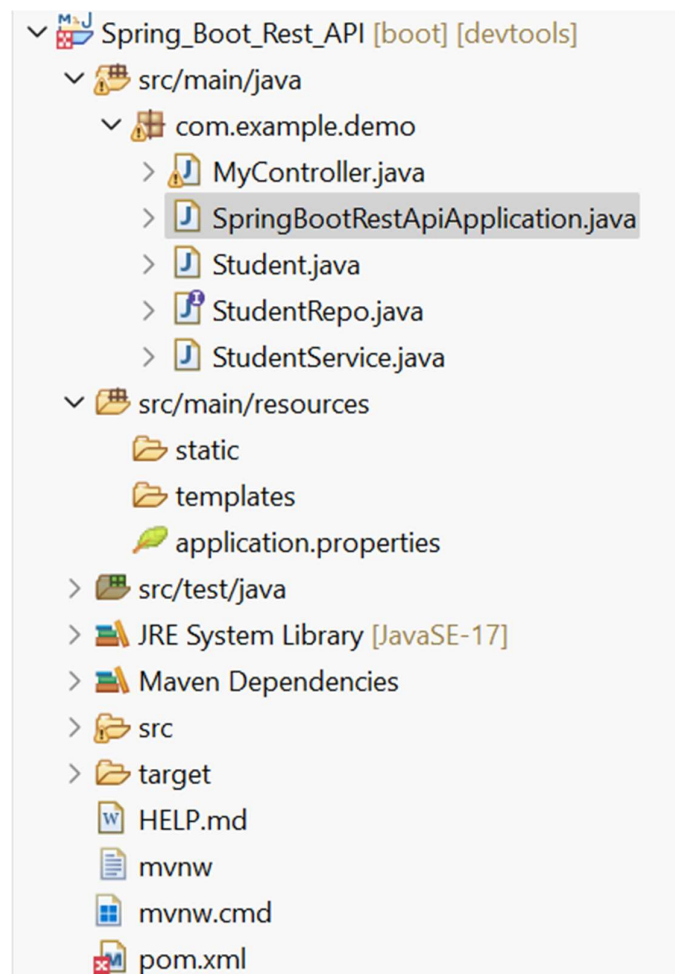
- **REST API:** A way to communicate between client and server using HTTP.
- **Stateless:** Every request is independent.
- **Resource-based:** Data = resources accessed via URLs.
- **Uses HTTP Methods:** GET, POST, PUT, DELETE.
- **Uses JSON:** For sending data.
- **Simple, scalable, fast.**

Spring Boot + REST API

Project Description :-

- This project is a **Spring Boot REST API** for handling **Student Registration**.
- It allows a client (Postman, React, Android, etc.) to **send student details**, and the API will store student information in the **database using Spring Data JPA**.

Project Structure :-



Step 1 :- pom.xml

```
https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
4     <modelVersion>4.0.0</modelVersion>
5     <parent>
6         <groupId>org.springframework.boot</groupId>
7         <artifactId>spring-boot-starter-parent</artifactId>
8         <version>4.0.0</version>
9         <relativePath/> <!-- lookup parent from repository -->
10    </parent>
11    <groupId>com.example</groupId>
12    <artifactId>Spring_Boot_Rest_API</artifactId>
13    <version>0.0.1-SNAPSHOT</version>
14    <name>Spring_Boot_Rest_API</name>
15    <description>Demo project for Spring Boot</description>
16    <url/>
17    <licenses>
18        <license/>
19    </licenses>
20    <developers>
21        <developer/>
22    </developers>
23    <scm>
24        <connection/>
25        <developerConnection/>
26        <tag/>
27        <url/>
28    </scm>
29    <properties>
30        <java.version>17</java.version>
31    </properties>
32    <dependencies> Add Spring Boot Starters...
33        <dependency>
34            <groupId>org.springframework.boot</groupId>
35            <artifactId>spring-boot-starter-webmvc</artifactId>
36        </dependency>
37
38        <dependency>
39            <groupId>org.springframework.boot</groupId>
40            <artifactId>spring-boot-devtools</artifactId>
41            <scope>runtime</scope>
42            <optional>true</optional>
43        </dependency>
44        <dependency>
45            <groupId>org.springframework.boot</groupId>
46            <artifactId>spring-boot-starter-webmvc-test</artifactId>
47            <scope>test</scope>
48        </dependency>
49
50
51        <dependency>
52            <groupId>org.springframework.boot</groupId>
53            <artifactId>spring-boot-starter</artifactId>
54        </dependency>
55
56        <dependency>
57            <groupId>org.springframework.boot</groupId>
```

```
58     <artifactId>spring-boot-devtools</artifactId>
59     <scope>runtime</scope>
60     <optional>true</optional>
61 </dependency>
62
63 <dependency>
64     <groupId>org.springframework.boot</groupId>
65     <artifactId>spring-boot-starter-test</artifactId>
66     <scope>test</scope>
67 </dependency>
68
69 <!-- New Added -->
70 <dependency>
71     <groupId>org.springframework.boot</groupId>
72     <artifactId>spring-boot-starter-web</artifactId>
73 </dependency>
74
75 <dependency>
76     <groupId>org.springframework.boot</groupId>
77     <artifactId>spring-boot-starter-data-jpa</artifactId>
78 </dependency>
79
80 <dependency>
81     <groupId>com.mysql</groupId>
82     <artifactId>mysql-connector-j</artifactId>
83 </dependency>
84
85 </dependencies>
86
87 <build>
88     <plugins>
89         <plugin>
90             <groupId>org.springframework.boot</groupId>
91             <artifactId>spring-boot-maven-plugin</artifactId>
92         </plugin>
93     </plugins>
94 </build>
95
96 </project>
```

Step 2 :- SpringBootRestApiApplication.java

```
1 package com.example.demo;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6 @SpringBootApplication
7 public class SpringBootRestApiApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(SpringBootRestApiApplication.class, args);
11     }
12
13 }
```

Step 3 :- Student.java

```
1 package com.example.demo;
2
3 import jakarta.persistence.Column;
4 import jakarta.persistence.Entity;
5 import jakarta.persistence.GeneratedValue;
6 import jakarta.persistence.GenerationType;
7 import jakarta.persistence.Id;
8 import jakarta.persistence.Table;
9
10 @Entity //Table Create
11 @Table(name="student") //Table Name
12 public class Student {
13
14     @Id
15     @GeneratedValue(strategy = GenerationType.IDENTITY) //Autoincrement
16     private Long id;
17
18     @Column //Create Column
19     private String fullname;
20
21     @Column(unique=true)
22     private String email;
23
24     @Column
25     private String password;
26
27     @Column
28     private String address;
29
30     public Long getId() {
```

```

31         return id;
32     }
33
34     public void setId(Long id) {
35         this.id = id;
36     }
37
38     public String getFullname() {
39         return fullname;
40     }
41
42     public void setFullname(String fullname) {
43         this.fullname = fullname;
44     }
45
46     public String getEmail() {
47         return email;
48     }
49
50     public void setEmail(String email) {
51         this.email = email;
52     }
53
54     public String getPassword() {
55         return password;
56     }
57
58     public void setPassword(String password) {
59         this.password = password;
60     }
61
62     public String getAddress() {
63         return address;
64     }
65
66     public void setAddress(String address) {
67         this.address = address;
68     }
69
70     @Override
71     public String toString() {
72         return "Student [id=" + id + ", fullname=" + fullname + ", email=" + email + ", password=" + password
73             + ", address=" + address + "]";
74     }
75 }

```


Step 4 :- StudentRepo.java

```
1 package com.example.demo;
2
3 import org.springframework.data.jpa.repository.JpaRepository;
4 import org.springframework.stereotype.Repository;
5
6 @Repository
7 public interface StudentRepo extends JpaRepository<Student, Long>{
8
9
10 }
```

Step 5 :- StudentService.java

```
1 package com.example.demo;
2
3 import java.util.List;
4
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.stereotype.Service;
7
8 @Service
9 public class StudentService {
10
11     @Autowired
12     StudentRepo sr;
13
14     public void registration(Student s)
15     {
16         sr.save(s); //Insert Query Directly
17     }
18
19     public List<Student> getallstudents()
20     {
21         return sr.findAll(); //select * from student
22     }
23 }
```


Step 6 :- MyController.java

```
1 package com.example.demo;
2
3 import org.springframework.beans.factory.annotation.Autowired;
4 import org.springframework.http.HttpStatus;
5 import org.springframework.http.ResponseEntity;
6 import org.springframework.web.bind.annotation.CrossOrigin;
7 import org.springframework.web.bind.annotation.GetMapping;
8 import org.springframework.web.bind.annotation.PostMapping;
9 import org.springframework.web.bind.annotation.RequestBody;
10 import org.springframework.web.bind.annotation.RestController;
11
12 @RestController
13 @CrossOrigin("*")
14 public class MyController {
15
16     @GetMapping("/")
17     public ResponseEntity<?> test()
18     {
19         return ResponseEntity.status(HttpStatus.ACCEPTED).body("Tested Ok");
20     }
21
22     @Autowired
23     StudentService ss;
24
25     @PostMapping("/register")
26     public ResponseEntity<?> registration(@RequestBody Student s)
27     {
28         try
29         {
30             ss.registration(s);
31             return ResponseEntity.status(HttpStatus.ACCEPTED).body("Registration Successfully Completed");
32         }
33         catch (Exception e)
34         {
35             return ResponseEntity.status(HttpStatus.BAD_REQUEST).body("Fail");
36         }
37     }
38 }
```

Step 7 :- application.properties

```
1 spring.application.name=Spring_Boot_Rest_API
2
3 spring.datasource.url=jdbc:mysql://localhost:3309/springbootdb1
4 spring.datasource.username=root
5 spring.datasource.password=
6
7 spring.jpa.hibernate.ddl-auto=update
8 spring.jpa.show-sql=true
9 spring.jpa.properties.hibernate.format_sql=true
```

```

:: Spring Boot :: (v4.0.0)

2025-12-09T15:33:20.124+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] c.e.demo.SpringBootRestApiApplication
2025-12-09T15:33:20.133+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] c.e.demo.SpringBootRestApiApplication
2025-12-09T15:33:20.253+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor
2025-12-09T15:33:20.254+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor
2025-12-09T15:33:21.570+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] s.d.r.c.RepositoryConfigurationDelegate
2025-12-09T15:33:21.681+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] s.d.r.c.RepositoryConfigurationDelegate
2025-12-09T15:33:22.693+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] o.s.boot.tomcat.TomcatWebServer
2025-12-09T15:33:22.723+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] o.apache.catalina.core.StandardService
2025-12-09T15:33:22.724+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] o.apache.catalina.core.StandardEngine
2025-12-09T15:33:22.840+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] b.w.c.s.WebApplicationContextInitializer
2025-12-09T15:33:23.191+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] o.hibernate.jpa.internal.util.LogHelper
2025-12-09T15:33:23.366+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] org.hibernate.Version
2025-12-09T15:33:24.552+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] o.s.o.j.p.SpringPersistenceUnitInfo
2025-12-09T15:33:24.637+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] com.zaxxer.hikari.HikariDataSource
2025-12-09T15:33:24.967+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] com.zaxxer.hikari.pool.HikariPool
2025-12-09T15:33:24.971+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] com.zaxxer.hikari.HikariDataSource
2025-12-09T15:33:25.089+05:30 WARN 29128 --- [Spring_Boot_Rest_API] [ restartedMain] org.hibernate.dialect.Dialect
2025-12-09T15:33:25.143+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] org.hibernate.orm.connections.pooling

Database JDBC URL [jdbc:mysql://localhost:3309/springbootdb]
Database driver: MySQL Connector/J

Database dialect: MySQLDialect
Database version: 5.5.5
Default catalog/schema: springbootdb1/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
JDBC fetch size: none
Pool: DatasourceConnectionProviderImpl
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown

2025-12-09T15:37:38.357+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] o.h.e.t.j.p.i.JtaPlatformInitiator
2025-12-09T15:37:38.394+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] j.LocalContainerEntityManagerFactoryBean
2025-12-09T15:37:38.511+05:30 WARN 29128 --- [Spring_Boot_Rest_API] [ restartedMain] JpaBaseConfiguration$JpaWebConfiguration
2025-12-09T15:37:38.767+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] o.s.boot.tomcat.TomcatWebServer
2025-12-09T15:37:38.775+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] c.e.demo.SpringBootRestApiApplication
2025-12-09T15:37:38.778+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [ restartedMain] .ConditionEvaluationDeltaLoggingListener
2025-12-09T15:38:06.301+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [nio-8080-exec-2] o.a.c.c.C.[Tomcat].[localhost].[/]
2025-12-09T15:38:06.301+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet
2025-12-09T15:38:06.303+05:30 INFO 29128 --- [Spring_Boot_Rest_API] [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet

Hibernate:
insert
into
student
(address, email, fullname, password)
values
(?, ?, ?, ?)

2025-12-09T15:38:06.318+05:30 WARN 29128 --- [Spring_Boot_Rest_API] [nio-8080-exec-2] org.hibernate.orm.jdbc.error
2025-12-09T15:38:06.319+05:30 WARN 29128 --- [Spring_Boot_Rest_API] [nio-8080-exec-2] org.hibernate.orm.jdbc.error

Hibernate:
insert
into
student
(address, email, fullname, password)
values
(?, ?, ?, ?)

2025-12-09T15:38:10.018+05:30 WARN 29128 --- [Spring_Boot_Rest_API] [nio-8080-exec-6] org.hibernate.orm.jdbc.error
2025-12-09T15:38:10.019+05:30 WARN 29128 --- [Spring_Boot_Rest_API] [nio-8080-exec-6] org.hibernate.orm.jdbc.error

Hibernate:
insert
into
student
(address, email, fullname, password)
values
(?, ?, ?, ?)

2025-12-09T15:38:12.265+05:30 WARN 29128 --- [Spring_Boot_Rest_API] [nio-8080-exec-8] org.hibernate.orm.jdbc.error
2025-12-09T15:38:12.265+05:30 WARN 29128 --- [Spring_Boot_Rest_API] [nio-8080-exec-8] org.hibernate.orm.jdbc.error

Hibernate:
insert
into
student
(address, email, fullname, password)
values
(?, ?, ?, ?)

2025-12-09T15:38:14.209+05:30 WARN 29128 --- [Spring_Boot_Rest_API] [io-8080-exec-10] org.hibernate.orm.jdbc.error
2025-12-09T15:38:14.210+05:30 WARN 29128 --- [Spring_Boot_Rest_API] [io-8080-exec-10] org.hibernate.orm.jdbc.error

Hibernate:
insert
into
student
(address, email, fullname, password)
values

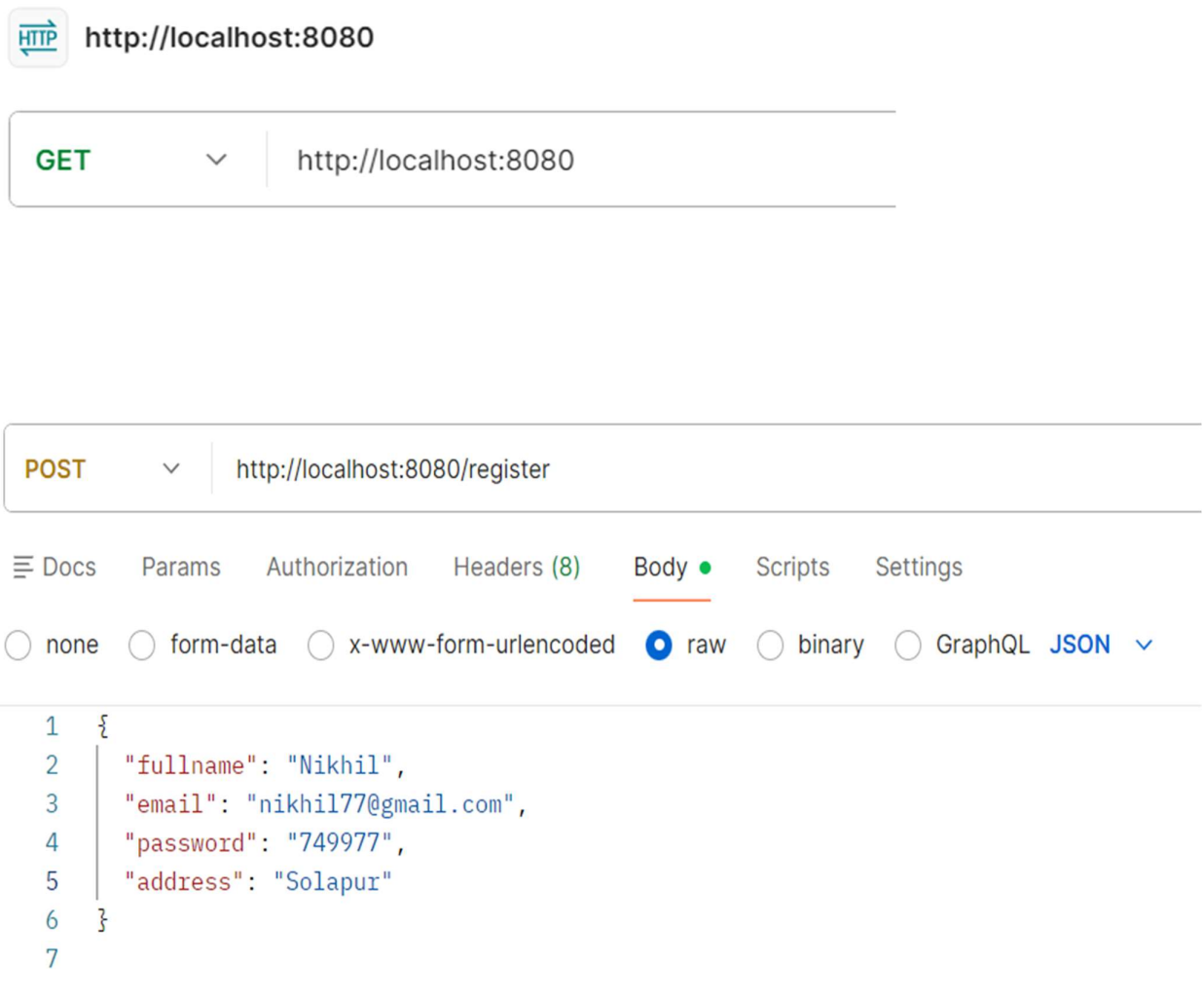
```

Server Output :-

The app runs at <http://localhost:8080>

Tested Ok

Postman :-



Body Cookies Headers (8) Test Results | ↺

 Raw ▾  Preview  Visualize ▾

1 Registration Successfully Completed

Database :-

id	address	email	fullname	password
3	Solapur	abc@gmail.com	abc	56789
4	Pune	xyz@gmail.com	xyz	749976
6	Pune	nikhil1234@gmail.com	Nikhil 1	123456
18	pune	abc2121@gmail.com	abcd xyz	12345
21	Solapur	nikhil77@gmail.com	Nikhil	749977